https://github.com/Harsh-IIT-Ropar/insurance-cost-prediction

# medical-insurance presentation

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# Problem and data

*This dataset is a realistic synthetic healthcare insurance dataset created for machine learning, data science, healthcare analytics, and AI-based prediction projects.*
*It contains demographic, lifestyle, medical history, chronic disease, hospitalization, and insurance-related information of individuals along with their annual medical costs.*

*The dataset is specially designed for:*

- *Medical Insurance Cost Prediction*
- *Healthcare Analytics*
- *Risk Assessment Systems*
- *Regression Machine Learning Projects*
- *Predictive Modeling*
- *AI Healthcare Dashboards*
- *Data Visualization & EDA*
- *Feature Engineering Practice*

📊 *Dataset Overview*

- *Rows: 7,500*
- *Columns: 22*
- *Target Variable: annual_medical_cost_usd*
- *Dataset Type: Structured Tabular Data*
- *ML Problem Type: Regression*

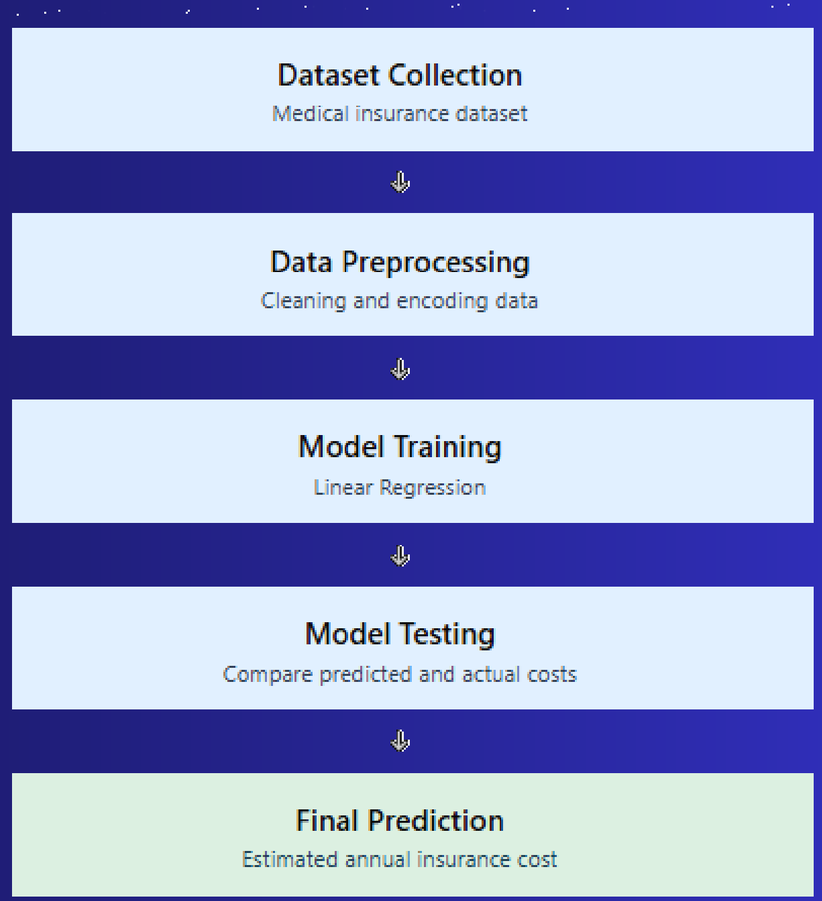
* *Approach Used – Linear Regression*

*Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values.*
*In our project, it predicts annual medical insurance charges based on different input factors such as age, BMI, smoking habits and other health-related details.*

* libraries used

1. numpy
2. pandas
3. matplotlib
4. scikit-learn
5. seaborn

# Project Pipeline Diagram

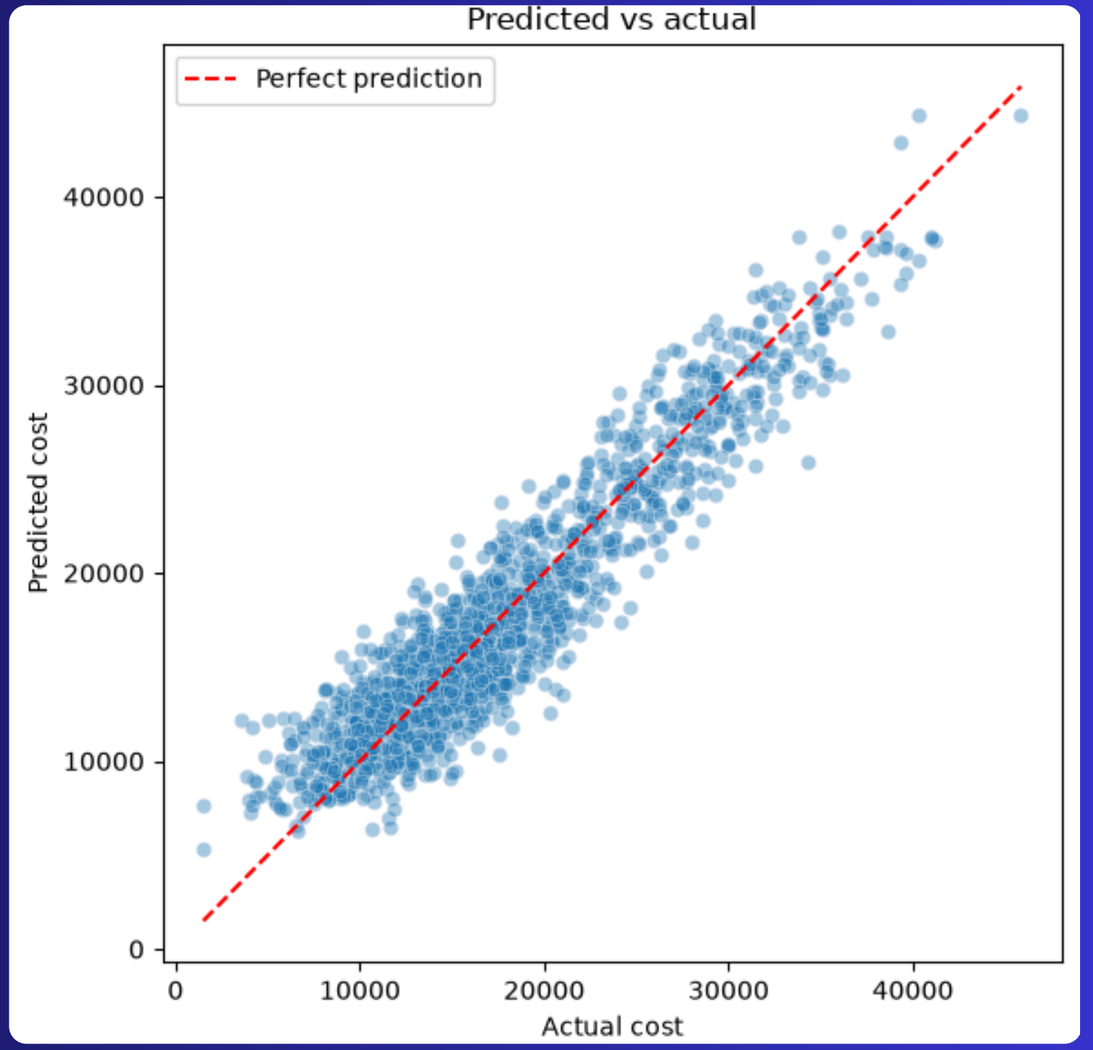


# Result of the Project

* After training and testing the Linear Regression model, we obtain predicted medical insurance costs based on the given input information.

Important Performance Metrics
We evaluate the model using the following metrics:

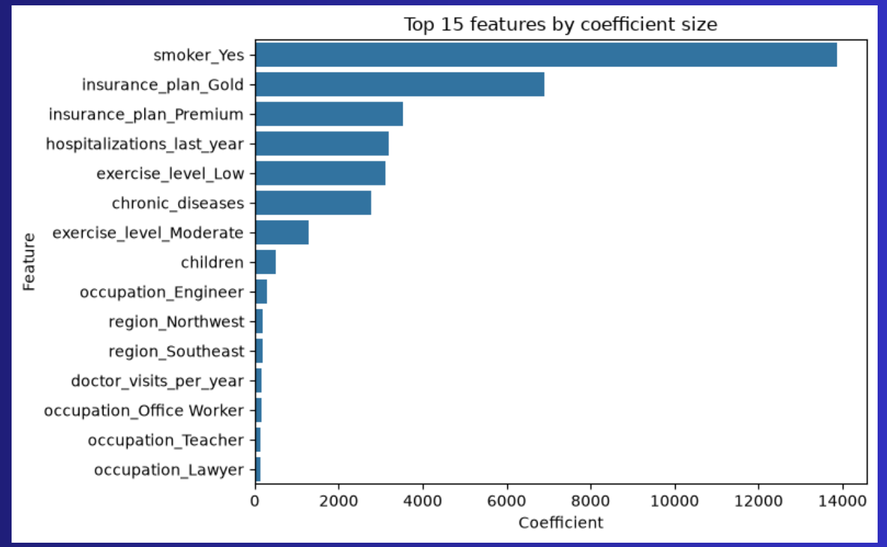
- The model explains about 88.5% of the variation in annual medical cost on the test set, with a typical error of about $2,000 to $2,500.
- Smoking status had the largest effect: holding other factors equal, smokers had a predicted cost about $13,900 higher than non-smokers.
- Chronic diseases, hospitalizations, the Gold insurance plan and low exercise level were the next strongest factors, while region and occupation had little effect.




*

| Metric | | Value |
|---|---|---|
| $R^2$ (test set) | - | 0.8852 |
| RMSE | - | 2516.85 |
| MAE | - | 2019.69 |

# Learnings




## 1. What Worked?

- Linear Regression successfully predicts numerical insurance costs.
- Personal and lifestyle information helps estimate insurance expenses.
- Pandas made data processing easier.
- Matplotlib helped visualize the results.

## 2. What Didn't Work?

- Prediction errors may occur because of differences between actual and predicted values.
- Linear Regression may not capture all complex relationships in medical insurance costs.
- Missing or incorrect data can affect model performance.

## 3. AI- Use Disclosure

AI tools were used to support project understanding, code assistance and presentation preparation. The dataset processing, model implementation and result analysis were performed as part of our project work.

# *thank you*

*Building a Healthier Tomorrow*